



Original Research Article

Plant reintroduction in restored peatlands: 80% successfully transferred – Does the remaining 20% matter?



Sandrine Hugron^{**}, Mélina Guéné-Nanchen, Noémie Roux,
Marie-Claire LeBlanc, Line Rochefort^{*}

Peatland Ecology Research Group, Centre for Northern Studies, Plant Science Department, Université Laval, 2425 rue de l'Agriculture, Québec, G1V 0A6, Canada

ARTICLE INFO

Article history:

Received 4 October 2019

Received in revised form 29 February 2020

Accepted 1 March 2020

Keywords:

Bogs

Ecological restoration

Moss layer transfer technique

Diaspore transfer

Plant material transfer

Reintroduction failures

ABSTRACT

In ecological restoration, especially in projects aiming at introducing a diverse plant community, species that do not establish are rarely reported. Yet, knowledge gained from identifying and understanding the reasons why species did not establish is useful to improve restoration techniques. In this paper, we used *Sphagnum* dominated peatlands restored with the Moss Layer Transfer Technique (MLTT) as a typical example of restored ecosystems for which information about species that does not establish is lacking. By comparing the plant species pools of 22 pairs of donor/restored *Sphagnum* peatlands (including 17 sites restored for more than 10 years), we calculated that the proportion of successfully introduced plant species (transfer rate) was $82 \pm 9\%$, amongst the highest transfer rate when compared to other restoration projects. Only five vascular plant species were classified as recalcitrant – species present in the donor site but absent from the restored sites: *Carex trisperma*, *Cypripedium acaule*, *Ilex mucronata*, *Maianthemum trifolium* and *Rubus chamaemorus*. Explanations about the filters impeding establishment were mainly based on the autoecology of the species. Depending on restoration goals (e.g. return of edible berries or orchid diversity) we recommend the specific reintroduction of recalcitrant species after the MLTT.

© 2020 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Ecological restoration is defined as the “process of assisting the recovery of an ecosystem that has been degraded, damaged or destroyed” (SERI, 2004). It is often considered as a tool for conservation (Aronson et al., 2006; Benayas et al., 2008; Possingham et al., 2015; Wiens and Hobbs, 2015; McDonald et al., 2016). Restoring biodiversity is frequently among the goals of restoration projects (SERI, 2004), especially since the enhancement of biodiversity was shown to be correlated with an increase in provision of ecological services (Benayas et al., 2008). Indeed, the effect of biodiversity on ecosystem functioning has been demonstrated in several studies, but especially for primary producers (Loreau et al.,

^{*} Corresponding author. Peatland Ecology Research Group, Centre for Northern Studies, Plant Science Department, Université Laval, 2425 rue de l'Agriculture, Québec, G1V 0A6, Canada.

^{**} Corresponding author. Peatland Ecology Research Group, Centre for Northern Studies, Plant Science Department, Université Laval, 2425 rue de l'Agriculture, Québec, G1V 0A6, Canada.

E-mail addresses: sandrine.hogue-hugron@fsaa.ulaval.ca (S. Hugron), melina.guene-nanchen.1@ulaval.ca (M. Guéné-Nanchen), noemie.roux1@gmail.com (N. Roux), marie-claire.leblanc@fsaa.ulaval.ca (M.-C. LeBlanc), line.rochefort@fsaa.ulaval.ca (L. Rochefort).

2001; Tilman et al., 2014). Plant diversity recovery in restoration projects sometimes becomes a proxy for the return of ecosystem functions (Couwenberg et al., 2011) and therefore, the evaluation of restoration success often focusses on plant diversity.

Plant reintroduction is a common strategy for promoting plant diversity in restored sites and shortcoming dispersal constraints (Meli et al., 2014; Perring et al., 2015; van Katwijk et al., 2016). When evaluating the effectiveness of reintroduction measures, negative results are generally published in grey literature (Godefroid et al., 2011) while scientific publications are biased towards positive results (Deredec and Courchamp, 2007; Godefroid et al., 2011; Fanelli, 2012). It is especially the case for reintroduction techniques focussing on the transfer of diverse plant communities such as substrate and/or diaspore transfer, compared to techniques aiming at introducing a few targeted species through sowing, planting or transplanting. For projects aiming at reintroducing diverse plant communities, success is mainly assessed by calculating transfer rates – the proportion of species established in the restored site in relation to the number of species contained in the transferred plant material (Hedberg and Kotowski, 2010; Kiehl et al., 2010; Török et al., 2011; Wolters et al., 2017). Species that do not establish are only rarely mentioned and reasons to explain those failures of establishment and subsequent colonisation are often not discussed (Hölzel and Otte, 2003; Donath et al., 2007; Koch, 2007; Coiffait-Gombault et al., 2011).

Literature about *Sphagnum* dominated peatland restoration using the Moss Layer Transfer Technique (MLTT) is a good example of the lack of analysis so far on the vascular plant that doesn't establish as the technique was first designed to favour peat moss establishment. The efficiency of the technique to re-establish a typical peatland vegetation cover was thoroughly evaluated by comparing the plant species pool in the restored sites with that of a regional reference ecosystem (D'Astous et al., 2013; Poulin et al., 2013; Rochefort et al., 2013). It was done by examining trajectories or drivers of plant communities establishing in over 50 different restoration projects (González et al., 2013; González and Rochefort, 2014) and by evaluating the return of peatland functions like carbon sequestration (Nugent et al., 2018; Strack and Zuback, 2013), hydrology (McCarter and Price, 2015) and resilience after fire (Blieher-Langdeau, 2019). Only a few studies (Poulin et al., 2011; D'Astous, 2012; Karofeld et al., 2016) compared the species pool between the peatland where the vegetation was collected, later called donor site, and the restored site. The goal of this comparison is to identify the recalcitrant plant species – the species present in the donor plant material, but that failed to establish in the restored site (Koch, 2007). From the previously cited studies, *Andromeda glaucophylla*, *Carex limosa*, *Eriophorum virginicum*, *Kalmia angustifolia*, *Rhynchospora alba*, *Rubus chamaemorus* and *Picea mariana* were identified as potentially recalcitrant to the MLTT in North America and *Drosera anglica* and *Rhododendron tomentosum* in Europe. We consider these species as “potentially recalcitrant” because all those studies examined only one restoration project and two out of three were short-term studies (conducted less than three years after the restoration). Acknowledging that: 1) three different outcomes of plant communities do usually establish following the application of the MLTT (González and Rochefort, 2019); 2) the plant communities in restored sites are dynamic and can change from one plant community to another as time since restoration increases (González and Rochefort, 2014); and 3) management decisions like density of donor plant material, delay between plant collection and reintroduction, efficiency of rewetting or introduction of a nurse crop influence the outcome of the restoration (González and Rochefort, 2014; Karofeld et al., 2016; Caporn et al., 2018); it is thus too risky to identify the missing species based on data from only one restoration project, especially if the time elapsed since the restoration is short.

Identifying the vascular plants failing to establish is important for improving the MLTT, if the goal is restoring the whole plant diversity of peatland project. To be accurate, identification of the lack of establishment should be based on data from multiple restoration projects and a minimum of 5 years post-restoration (González and Rochefort, 2014). The objective of this study is thus to determine 1) the efficiency of the MLTT in restoring plant biodiversity (i.e. transfer rate), 2) if there are any species recalcitrant to the MLTT, 3) the possible reasons explaining why those species are missing, and 4) suggest improvements to the MLTT to favour the establishment of those species.

2. Material and methods

2.1. Description of the MLTT

In Canada, *Sphagnum* peatlands restoration after peat extraction is performed with the MLTT. The method consists of 1) reprofiling the topography of the cutover site to flatten the surface and ensure an even distribution of water (by creating terraces and bunds, depending on the landscape settings), 2) collecting plant material (top 10 cm of the moss carpet) in a donor peatland, 3) spreading the plant material over the restoration site in a 1:10 to 1:15 ratio (1 m² of plant material is spread over 10–15 m² of the restoration site), 4) covering the plant material with a straw mulch for protection, 5) fertilizing with a low dose of phosphorus (P₂O₅; 19.5 kg/ha) and 6) blocking the drainage ditches to raise water level (see Graf and Rochefort, 2016 for a description of the conceptual approach and Quinty and Rochefort, 2003 for the technical details). As of 2015, more than 100 restoration projects had been implemented across Canada, restoring a total area of 1085 ha (CSPMA, 2015). The majority of these projects were implemented entirely by the Canadian Horticultural Peat Industry. Since 1998, all those sites are systematically surveyed by the Peatland Ecology Research Group (PERG) 3, 5, 7, 10, 15, 20 and 25 years post-restoration. The surveys are performed in permanent plots in which the cover of each plant species is evaluated systematically (see González and Rochefort (2014) for a detailed description of the method).

2.2. Site selection

For this study, we selected restoration projects for which the donor site composition was either known from previous studies or was still existing for field surveys. Among those, only restored sites aged 4 years and more (although the majority were older than 10 years), presenting an average *Sphagnum* cover of 45% and exhibiting less than 10% of bare peat cover were chosen. The goal of choosing sites where the restoration was judged successful was to remove the bias of choosing a site where the restoration technique was not properly applied. In total, 22 pairs (donor/restored sites) were selected, all located in Eastern Canada (Fig. 1; Table 1).

2.3. Transfer rate

To calculate the transfer rate, we built a database of all species (including bryophytes and liverworts) present in 21 of the 22 donor sites and their associated restored sites (excluding Baie-Ste-Anne, for which the donor site bryophyte and liverworts data was not available). To do so, additional field surveys were performed and data from the PERG long-term database of vegetation present in restored sites and from Guéné-Nanchen et al. (2018) were used. We then compared the number of plant species present in the donor sites with the ones found in their associated restored sites.

2.4. Recalcitrant vascular plant species

2.4.1. List of potentially recalcitrant species

A list of potentially recalcitrant vascular plant species was established by comparing the species surveyed in a restored site with its associated donor site. At this step, two cases could occur: 1) species occurring both in the donor and restored sites: the species was considered common and not added to the list or 2) species occurring in the donor site but not found in its associated restored site: the species was added to the list. Because donor site surveys were all performed after plant material collection, and that Guéné-Nanchen et al. (2018) concluded that vegetation cover was not 100% representative of the original vegetation before plant collection, we also compared the vegetation in the restored sites with vegetation surveys performed in regional reference sites (Fig. 1). Once again, any species present in the reference site, but absent in its associated restored site was added to the list. A total of 42 vascular plant species were classified as potentially recalcitrant (column 1 of Table 2). Bryophytes and lichens were not included in this part of the study because of time constraints and logistical considerations, but will be considered in a future study.

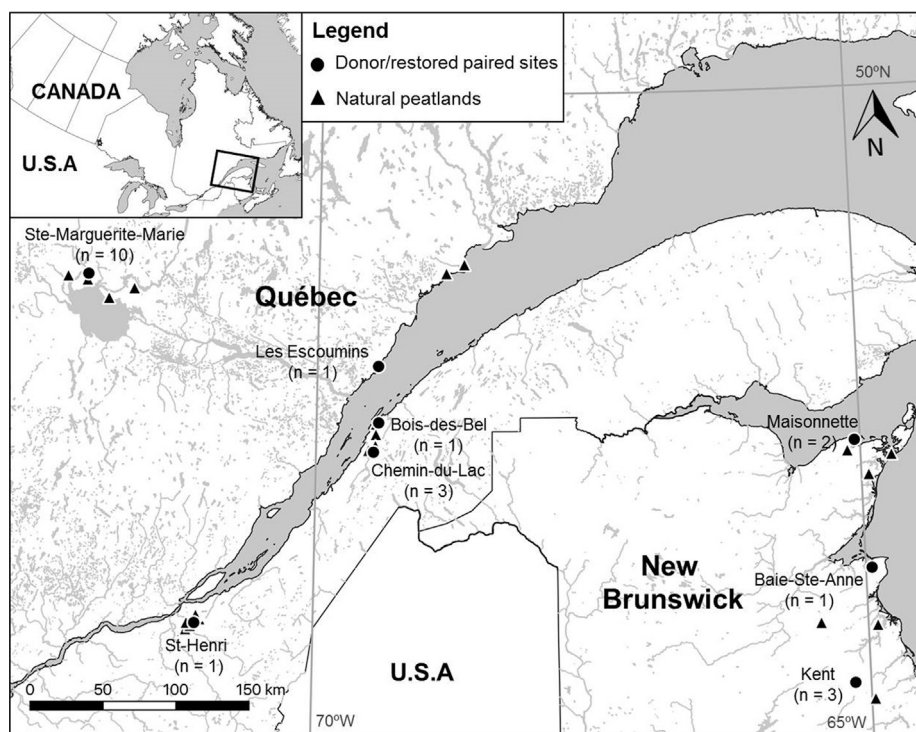


Fig. 1. Location of the 22 pairs of donor/restored sites and the 27 natural peatlands in the province of Québec and New Brunswick (Eastern Canada) included in this study.

Table 1

List of the studied restored sites and their associated donor sites.

Restored sites (n = 22)			Associated donor sites (n = 12)
Name of peatland (total area)	Restoration project	Years since restoration	Surface restored (ha)
St-Henri (100 ha)			St-Henri
Project A	4	9	
Bois-des-Bel (100 ha)			St-Antonin A
Project A	17	12	
Chemin-du-Lac (1500 ha)			St-Antonin B
Project A	15	3	
Project B	14	5	St-Antonin C
Project C	13	11	St-Antonin D
Ste-Marguerite-Marie (6500 ha)			Ste-Marguerite-Marie A-J ^a
Project A	17	1	
Project B	16	15	
Project C	16	10	
Project D	16	11	
Project E	16	10	
Project F	16	10	
Project G	15	17	
Project H	15	17	
Project I	14	12	
Project J	14	15	
Les Escoumins (800 ha)			Les Escoumins
Project A	5	12	
Baie-Ste-Anne (7200 ha)			Baie-Se-Anne
Project A	6	2	
Kent (600 ha)			Kent A
Project A	15	5	
Project B	9	7	
Project C	9	4	Kent 2008 B–C ^a
Maisonnette (400 ha)			Maisonnette A–B ^a
Project A	16	11	
Project B	10	9	

^a Limits of the donor sites for each individual restoration project was not possible to determine. One vegetation survey was performed on the whole area of the donor site.

2.4.2. Field surveys

In order to confirm the absence of the recalcitrant species, additional field surveys were conducted in all restored sites. The potentially recalcitrant species were sought by walking systematically the sites. Because timespan for vegetation surveys was limited from mid-June to mid-July (the blooming period of orchids and *Carex*) and because vegetation composition of restored sites is known to be homogeneous (excluding ditches; Fontaine et al., 2007), the area surveyed was proportional to the size of the site. In the donor sites, the entire area where plant collection occurred was surveyed following the technique previously described. In large donor sites where restoration material was collected for different years or multiple restoration project, it was sometimes difficult to determine *a posteriori* the boundary of the donor area collected for each restoration project. In these cases, the entire area of the donor site was surveyed without trying to further subdivide in between restoration projects (Table 1).

2.4.3. Management and environmental variables

To determine why certain species were recalcitrant in some restored sites and not in others, we compiled different explanatory variables (four categories: management, peat chemistry, meteorology and hydrology). Management variables comprised: 1) number of growing seasons following the restoration operations, 2) season when the restoration was performed, 3) application of phosphate (yes, in part, no), 4) delay of restoration, defined as the number of months between plant collection and spreading. A composite peat sample was collected in each restored sector the third or fifth year after the restoration. A subsample saturated with deionized water (Soil Medium Extract method) was used to measure pH (AB150 pH meter, Accumet) and electrical conductivity (A122 portable conductivity meter, Orion Star) which was corrected with Sjörs (1950) formula. Approximately 100 g of peat was sifted, dried at 40 °C and analyzed to evaluate the concentration of Ca, Mg, Fe, Mn, K, Na, available P and N/NH₄⁺. Calcium, Mg, Fe, Mn, K and Na were extracted according to Amacher et al. (1990) and analyzed with an ICP spectrometer (Optima 4300 DV, PerkinElmer). Available P was extracted following the method of Bray and Kurtz (1945) and analyzed with the ICP. Finally, N/NH₄⁺ was extracted according to Keeney and Nelson (1982) and analyzed with a Quikchem 8500 series 2 FIA (Lachat). Weather data from the first growing season following the restoration of each site were acquired from the closest meteorological station. Mean temperature and cumulative precipitation for spring, summer and fall were calculated following González and Rochefort (2014). The degree of blockage of the drainage ditches

Table 2

Classification of the 42 potential recalcitrant species considered in this study.

Species	Present in x% of:		% of projects in which species is recalcitrant
	Restored	Donor	
	sites (22 pairs)		
Recalcitrant Species			
<i>Carex trisperma</i>	36	64	57
<i>Cypripedium acaule</i>	14	36	88
<i>Ilex mucronata</i>	18	41	56
<i>Maianthemum trifolium</i>	9	86	89
<i>Rubus chamaemorus</i>	9	32	86
Common Peatland Species			
<i>Andromeda glaucophylla</i>	77	68	0
<i>Carex oligosperma</i>	77	68	0
<i>Drosera rotundifolia</i>	95	95	5
<i>Eriophorum angustifolium</i>	73	77	39
<i>Larix laricina</i>	95	91	5
<i>Picea mariana</i>	95	100	5
<i>Sarracenia purpurea</i>	91	100	10
<i>Vaccinium angustifolium</i>	77	45	0
Species Establishing Spontaneously		Dispersal vector	
<i>Aronia melanocarpa</i>	27	14	Animals
<i>Betula glandulosa</i>	45	5	Wind
<i>Eriophorum russeolum</i>	36	0	Wind
<i>Eriophorum virginicum</i>	41	23	Wind
Species Non-Targeted by the MLTT		Habitat	
<i>Acer rubrum</i>	14	23	Laggs
<i>Amelanchier bartramiana</i>	23	18	Laggs
<i>Betula michauxii</i>	9	0	Facultative of peatlands
<i>Calopogon tuberosus</i>	9	14	Wet habitats
<i>Carex exilis</i>	0	0	Wet habitats
<i>Carex limosa</i>	14	5	Wet habitats
<i>Carex pauciflora</i>	0	9	Wet habitats
<i>Coptis groenlandicum</i>	9	18	Laggs
<i>Drosera anglica</i>	0	0	Wet habitats
<i>Drosera intermedia</i>	14	5	Wet habitats
<i>Empetrum nigrum</i>	14	0	Facultative of peatlands
<i>Gaylussacia baccata</i>	23	18	Facultative of peatlands
<i>Geocaulon lividum</i>	9	0	Laggs
<i>Maianthemum canadense</i>	0	0	Facultative of peatlands
<i>Melampyrum lineare</i>	5	0	Laggs
<i>Menyanthes trifoliata</i>	0	0	Wet habitats
<i>Platanthera blephariglottis</i>	14	23	Undisturbed bogs
<i>Pogonia ophioglossoides</i>	14	23	Wet habitats
<i>Rhynchospora alba</i>	14	5	Wet habitats
<i>Salix discolor</i>	23	0	Facultative of peatlands
<i>Trichophorum cespitosum</i>	0	0	Wet habitats
<i>Utricularia cornuta</i>	5	5	Wet habitats
<i>Vaccinium macrocarpon</i>	23	0	Wet habitats
<i>Vaccinium myrtilloides</i>	18	5	Facultative of peatlands
<i>Xyris montana</i>	9	0	Wet habitats

between the peat fields was assessed on four levels (1 = completely open, 2 = collapsed <50%, 3 = collapsed > 50%, 4 = completely collapsed) for each restored sector and used as a hydrological proxy.

2.5. Data analyses

Species that were previously identified as potentially recalcitrant were classified using a decision tree into four groups: **Recalcitrant Species**, **Common Peatland Species**, **Species Establishing Spontaneously** and **Non-Targeted by the MLTT** (Fig. 2). The decision tree was built using criteria based on the proportion of donor/restored sites in which the potentially recalcitrant species were present. More specifically, for the species that were present in more than 25% of the donor sites, we made an assessment of the presence of each species in a restored site and its associated donor site to determine if the species was “recalcitrant” to this specific restoration project. If a species was “recalcitrant” in 50% or more of the restoration projects, it was classified as a Recalcitrant Species.

To determine if explanatory variables could explain the absence of **Recalcitrant Species** in restored sites, we first performed a (Spearman) correlation to verify if a link exists between the frequency of a given **Recalcitrant Species** in the restored

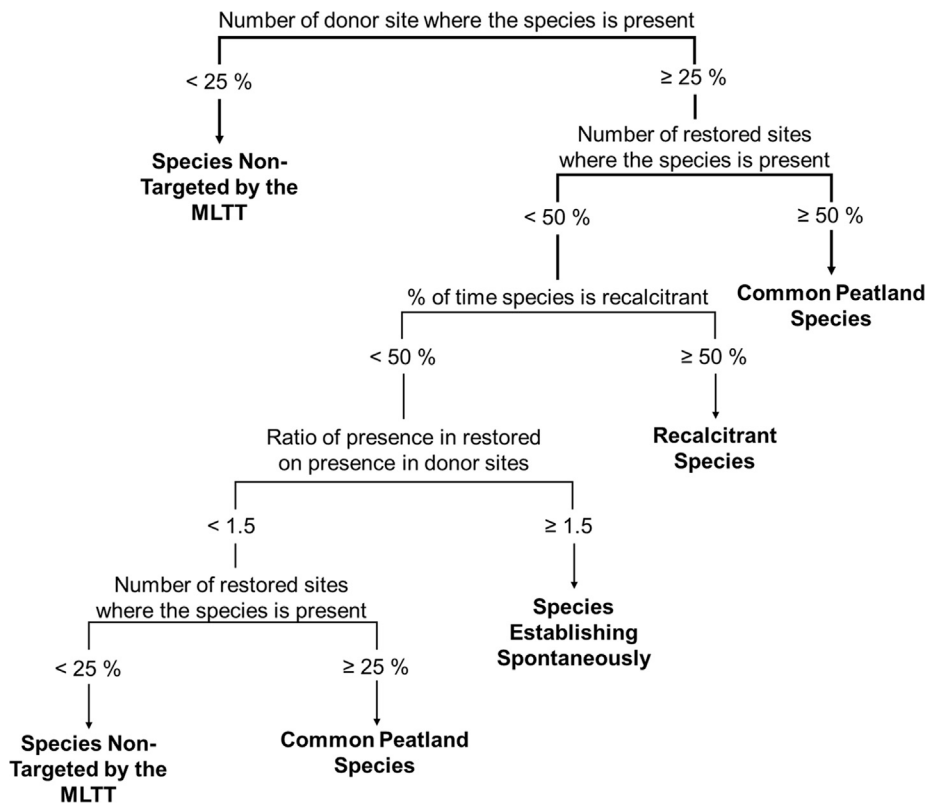


Fig. 2. Decision tree used to categorize each plant species.

sites and its associated donor sites. We then performed a redundancy analysis (RDA) to evaluate the role of management and environmental variables on the presence of recalcitrant species in restored sites. For this analysis, in addition to nutrient concentrations, nutrients ratio (N:P, N:K, K:P, Ca:Mg and K:Mg) were also considered. A binary matrix was used to represent whether a recalcitrant species was present or absent in the restored sites (dimensions: 20 x 5). We excluded sites for which peat samples were not collected (St-Henri, Les Escoumins). Because collinearity between explanatory variables was expected, a forward selection was implemented before running the RDA. Explanatory variables highly correlated with others (Spearman coefficient > 0.7) were also removed. The significance of the RDA was tested with a permutation test using 9999 randomized runs (Legendre and Legendre, 2012). All analyses were conducted with R software (version 3.3.1; R Core Team, 2017), more precisely with `decostand`, `ordistep` and `rda` from `vegan` (Oksanen et al., 2016) and `rcorr` from `Hmisc`.

3. Results

3.1. Transfer rate

On average, the 21 donor sites studied contained 24 ± 3 species (mean $\pm$ SD; min = 22; max = 32), including bryophytes. The mean transfer rate ($\pm$ SD) was estimated to $82 \pm 9\%$ (max = 100%; min = 71%).

3.2. Classification of vascular plant species

Of the 42 potentially recalcitrant species, only five were classified in the **Recalcitrant Species** group (Table 2). Three species exhibited a very high level of recalcitrance (recalcitrant in more than 85% of the restoration projects in which the species was transferred): *Cypripedium acaule*, *R. chamaemorus* and *Maianthemum trifolium*. The latest stands out as particularly unsuccessfully reintroduced by the MLTT: it was present in 86% of the donor sites (with a mean frequency $\pm$ SD of $26 \pm 19\%$), but was only present in 9% of the restored sites (with a mean frequency $\pm$ SD of $9 \pm 7\%$; Table 2).

Of the 42 species initially listed as potentially recalcitrant, eight species were classified as **Common Peatland Species** (Table 2). One must keep in mind, however, that many vascular plant species were readily excluded from this study at the database analysis step because they were always present in both the restored site and its associated donor site. Examples of these excluded species are *Chamaedaphne calyculata*, *Eriophorum vaginatum*, *Kalmia angustifolia*, *Kalmia polifolia*, *Rhododendron canadense*, *Rhododendron groenlandicum*, *Vaccinium oxycoccos* that were all identified by previous studies as

commonly observed in restored sites (D'Astous et al., 2013; Poulin et al., 2013; Rochefort et al., 2013; González and Rochefort, 2014; González and Rochefort, 2019).

Four species were classified in the **Species Establishing Spontaneously** group (Table 2), comprising species that occurred in low abundance (or absent) in donor sites but were found in an unexpectedly high proportion of restored sites. Most of these species are wind-dispersed (Flora of North America, 1993+).

Finally, more than half the species studied were classified as **Non-Targeted by the MLTT** because they were present in less than 25% of the donor site (Table 2). The restoration managers are specifically instructed to target donor sites composed of open *Sphagnum* carpets dominated by the subgenus *Acutifolia* with a presence of *Polytrichum strictum* (Quinty and Rochefort, 2003). Thus, species belonging to wetter habitats – i.e. pools and wet lawns – or habitats with closed canopy – i.e. treed islands or lags – are not directly transferred by the MLTT unless their seeds happen to be dispersed onto the donor site prior to plant collection.

3.3. Environmental/management causes of recalcitrance

The frequency of a given **Recalcitrant Species** in the restored sites was not correlated with its frequency in the associated donor sites, except for *I. mucronata* (Spearman $r^2 = 0.72$; $P > 0.001$). The four other species, *C. trisperma*, *C. acaule*, *M. trifolium* and *R. chamaemorus*, exhibited non-significant correlation. The peat Mg concentration was the only other variable explaining significantly ($F = 10.08$, $P = 0.005$) the presence of **Recalcitrant Species** in restored sites according to the RDA model (not shown, explaining 34% of the variability). Indeed, Mg concentration was at least three times higher in restored sites where *C. trisperma*, *I. mucronata*, *M. trifolium* and *R. chamaemorus* were present (Table 3). It should be noted, however, that *M. trifolium* and *R. chamaemorus* were only present in respectively one and two restored sites. Only *C. acaule* showed the reverse trend, being present in one restored site with Mg concentration lower than average (Table 3).

4. Discussion

4.1. Assessing success of the MLTT to restore plant diversity

This study confirms that the MLTT is an effective method to restore the plant diversity of *Sphagnum* peatlands following peat extraction activities. On average, 82% of the species targeted by the technique – i.e. the species present in the donor site and transferred into the site to be restored – establish successfully. This transfer rate stands among the highest transfer rates reported for restoration projects using a plant or soil transfer method, which is on average 68% (min: 8%, max: 90%) for wetland restoration projects and 62% (min: 0, max: 100) for dry grasslands (averages calculated from transfer rates reported by Kiehl et al. (2010) and Hedberg and Kotowski (2010)). Because vegetation surveys in donor sites were not performed prior to plant collection and because plant composition might have changed following the disturbance caused by collection (Guéné-Nanchen et al., 2018), the transfer rates we calculated are probably not entirely accurate for each site individually. We, however, think that the mean transfer rate calculated for the 21 restored sites is representative of the reality.

4.2. Identifying the species recalcitrant to the MLTT

Only five species were classified in the **Recalcitrant Species** group. We, however, acknowledge that the number of species classified in each category was influenced by thresholds that were established subjectively (Fig. 2). However, even when modifying the threshold value at which a species is considered recalcitrant, the increase of Recalcitrant Species would remain negligible. For example, if the threshold for presence in donor sites was lowered to 20% instead of 25%, four additional species

Table 3

Mean peat Mg concentration ($\text{mg g}^{-1} \pm \text{SE}$) in restored sites ($n = 19$; excluding St-Henri and Les Escoumins sites where chemistry samples were not taken) where the **Recalcitrant Species** were present or absent.

Recalcitrant Species	Presence/absence	n (restored sites)	Concentration (mg g^{-1})		
<i>Carex trisperma</i>	Present	7	0.96	$\pm$	0.23
	Absent	12	0.28	$\pm$	0.05
<i>Cyripedium acaule</i>	Present	1	0.23		
	Absent	18	0.55	$\pm$	0.12
<i>Ilex mucronata</i>	Present	4	1.17	$\pm$	0.29
	Absent	15	0.36	$\pm$	0.09
<i>Maianthemum trifolium</i>	Present	2	1.32	$\pm$	0.58
	Absent	17	0.44	$\pm$	0.10
<i>Rubus chamaemorus</i>	Present	1	1.9		
	Absent	18	0.45	$\pm$	0.10

would have been classified as **Recalcitrant Species**: *Acer rubrum*, *Eriophorum virginicum*, *Platanthera blephariglottis* and *Pogonia ophioglossoides*. Alternatively, we also could have lowered the threshold for the proportion of the sites where the species is recalcitrant. For example, decreasing the threshold to 25% (instead of 50%) would have resulted in only one additional species classified as **Recalcitrant Species**, *Eriophorum angustifolium*.

Out of those five species that could potentially have been classified as **Recalcitrant Species** if the thresholds had been set differently, we feel that at least two species belonged better to their initial groups. We think that *E. virginicum* was best classified as a species **Establishing Spontaneously** because 75% of the time the species was present in the restored sites, it was recruited from the surrounding environment (i.e. was not present in the donor site; data not shown). Since this species produces a large number of seeds that are wind-dispersed over long distances (Flora of North America, 1993+), it is not surprising that it easily colonizes restored sites. We also believed that the initial classification of *E. angustifolium* as a **Common Peatland Species** was more appropriate because its absence was restricted to 10 different restoration projects all located on one extensive peatland (Ste-Marguerite-Marie). For those 10 restoration projects, we were not able to define the boundaries between the individual donor sites that were all adjacent to one another. As a result, the 10 restoration projects were compared to the species pool found over the entire area of the donor sites combined. From the report on the restoration project by Quinty (2005) for this peatland, we were able to determine that prior to plant collection in this donor site, *E. angustifolium* was virtually absent (less than 1% cover). We thus believe that the recalcitrance for this species was overestimated. Overall, even when lowering both threshold values substantially, the number of species classified as **Recalcitrant Species** remains low compared to the number of species that successfully establish in restored sites, showing that this result is quite robust.

4.3. Explaining the presence or absence of recalcitrant species

Since none of the **Recalcitrant Species** were absent from all the restored sites, we hypothesized that some environmental characteristics of the restored site or management decisions could explain their occurrence. The concentration of Mg in the post-extraction residual peat was the only explanatory variable significantly correlated to the presence of **Recalcitrant Species** (albeit low abundance) in the restored sites. When present, *I. mucronata*, *R. chamaemorus*, *M. trifolium* and *C. trisperma* tended to be found in restored sites exhibiting high concentration of Mg. In Eastern Canada, the four species can generally be found in richer peatland habitats like poor minerotrophic peatlands or lags (Wheeler et al., 1983; Gignac et al., 2004; Paradis et al., 2015). Additionally, *I. mucronata* was also more frequently found in restored sites when it was transferred from a donor site exhibiting high abundance of the species (frequency of 10% or more). This species does not disperse easily over long distances, so sites where it is highly abundant enhanced the chances of transferring fruits within the donor material. Overall, the fact that only two factors could be identified for explaining the occurrence of **Recalcitrant Species** in restored sites indicates that their autoecology is probably playing a more important role than environmental conditions or management decisions.

For the five plant species classified as **Recalcitrant Species**, we were able to identify three reasons explaining their absence in the restored sites. *M. trifolium* and *R. chamaemorus* reproduce mainly by rhizomes so we believe that the fragmentation of the rhizome during the restoration process is problematic. Donor plant material is fragmented by a rotovator and a manure spreader during the steps of plant collection and introduction, resulting in fragments usually less than 2 cm long deposited on top of the peat. Such short sections of rhizomes are probably not sufficient for the regeneration of new individuals and establishment of the species in the restored sites. Indeed, when propagated vegetatively, rhizome sections of *R. chamaemorus* and of species of the genus *Maianthemum* are respectively 20 cm and 5–20 cm long and positioned inside the substrate, rather than deposited at the surface (Luna et al., 2008; Bellemare et al., 2009; Lady Bird Johnson Wildflower Center, 2014). For *C. trisperma* and *I. mucronata*, the relatively short time elapsed since the restoration (4–17 years in this study) was probably not sufficient for their recovery. Even if this variable was included in the RDA analysis but was not found significant, we think it exerts an indirect effect. In natural peatlands, the two species are generally found under the canopy of other plants (Wheeler et al., 1983; Gignac et al., 2004; Paradis et al., 2015). Indeed, Paradis and Rochefort (2012) showed that the germination rate of both species was 3–5 times higher in shaded conditions compared to full sunlight. The lack of such habitats in the relatively young restored sites probably explains why those species were classified as **Recalcitrant Species**. Concerning the remaining species classified as **Recalcitrant Species**, *C. acaule*, many factors of the complex biology of orchids could be responsible for its low occurrence in restored sites. In this study, the species exhibited a clear preference for older restored sites (10 years and more), which coincides with the known 8–10 years' delay between germination and first flowering of this species (Curtis, 1943). The species is also known to reproduce mainly from seeds (Curtis, 1943) whose germination and development is restricted by the presence of compatible mycorrhizal fungi to provide essential elements and nutrition (Shefferson et al., 2005; Bunch et al., 2013). However, even if no studies specifically tested if the mycorrhizal fungi associated with orchid germination was transferred by the MLTT, we do not believe this would be the main limiting factor for two reasons: 1) the MLTT was shown to be generally favorable to biomass and microbial activity (bacteria and fungi) in comparison with unrestored peat (Andersen et al., 2006), and 2) *Sphagnum* mosses collected in natural peatlands and placed in petri dishes were shown to be an adequate *ex situ* germination substrate for another *Sphagnum* bog orchid species, *P. blephariglottis* (Lemay et al., 2015).

4.4. Improving of the Moss Layer Transfer Technique

The main goal of peatland restoration in Canada is to return the carbon sequestration function (Graf and Rochefort, 2016), which was demonstrated to be restored within a decade to fifteen years (Nugent et al., 2018). Hence, we consider that little can be done in terms of modifying the MLTT to accommodate the recalcitrant species that would not be detrimental to a whole suite of other species or ecosystemic functions. For example, adding magnesium along with phosphorus during the fertilization step could probably improve the establishment conditions of *C. trisperma*, *I. mucronata*, *M. trifolium* and *R. chamaemorus*, but we do not recommend applying this amendment unless trials are performed previously to determine the optimal dose and verify its effect on *Sphagnum* mosses and other peatland vegetation. Even if fragmentation of the rhizome was identified as a possible cause for the poor establishment of *R. chamaemorus* and *M. trifolium* in restored peatland, we do not recommend modifying the restoration technique to reduce fragmentation. From our field experience, we learned that fragmentation of the moss carpet by the machinery is beneficial for the establishment of *Sphagnum* mosses because it ensures that plant material is spread in a uniform thin layer (~1 cm).

In the case where the return of plant diversity is an important goal in a given restoration project, additional actions could be performed after the restoration to favour the establishment of **Recalcitrant Species**. For example, planting trees and shrubs to accelerate the formation of a canopy could favour the germination of *C. trisperma* and *I. mucronata*. Another strategy could be to introduce targeted **Recalcitrant Species** a few years after restoration operations by sowing or planting. Of course, these additional steps would increase the cost of the restoration projects.

If the goal of a restoration project is to restore the plant diversity associated with natural peatlands, the introduction of plant species **Non-Targeted by the MLTT** could become an interesting option, as long as their habitats are present in the restored sites. For the plant species usually found in wet peatland habitats, it is possible to create shallow pools where wetter conditions are found throughout the year. Even when this type of habitat is created in restored peatlands, some vascular plant species associated with pool habitats exhibited very low establishment rates when the MLTT was applied (Poulin et al., 2011) and were more successfully introduced by sowing (Landry et al., 2012; Laberge et al., 2015). Lagg species usually need shaded conditions. Planting peatland trees to rapidly create those conditions at the margin of restoration projects could again contribute to their establishment. Restoring the connectivity between the forested margin and the restored site could contribute to the creation of lagg environment (Lefebvre-Ruel et al., 2019), thus creating a more suitable habitat for some of the species **Non-Targeted by the MLTT**.

5. Conclusions

Studies focussing on reintroduction failures are becoming more common in the scientific literature when the goal of the introduction is species recovery, which usually targets only one or few species with a special conservation status (Griffith et al., 1989; Deredec and Courschamp, 2007; Grey-Ross et al., 2009; Dalrymple et al., 2012; Bennett et al., 2013). To our knowledge, this study is one of the few specifically aiming to identify the reintroduction failures following the transfer of a diverse plant community for ecological restoration. Adding to its uniqueness, this study was performed on more than 20 different restoration projects and more than half of them were aged 10 years or more.

We showed that reintroduction failures – i.e. recalcitrant species – could be identified for any type of restored ecosystem by comparing the species pool in the restored sites with the one in its associated donor sites. This analysis should, however, be performed on a large number of sites restored with the same technique and time elapse since the restoration should be greater than five years. Ideally, vegetation surveys in the donor sites should be performed before plant material collection for restoration. In certain cases, it can be performed *a posteriori* if the donor site has the ability to regenerate swiftly after plant collection, which was shown to be the case for peatlands used as donor sites for restoration with the MLTT (Guéné-Nanchen et al., 2018). We also suggested that thresholds for classifying plant species need to be set with a certain part of subjectivity, taking into account hands-on field knowledge and the number of pairs of donor/restored sites. However, to ensure the robustness of the classification, simulations of threshold modifications should be performed and the fate of species switching groups should be addressed carefully.

Reintroduction failures are, overall, important to report for adaptive management. The knowledge gained from these failures is equally valuable for filling knowledge gaps as are reintroduction successes (Stravinsky, 2000; Dalrymple et al., 2012). Failed reintroduction can also indicate that monitoring was done at the wrong time scale (Godefroid et al., 2011), therefore suggesting a more adequate timespan for the evaluation of biodiversity recovery. Thereupon, Dalrymple et al. (2012) suggested that detailed monitoring should be pursued for more than 10 years after propagule reintroduction.

Funding source

This work was supported by the National Sciences and Engineering Research Council of Canada [grant number CRDP/437463-12]; MITACS [grant number IT08943]; and the Canadian Sphagnum Peat Moss Association and its members.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful to Berger, Lambert Peat Moss, Premier Tech Horticulture, Scotts Canada and Sun Gro Horticulture for site access. Special thanks goes to Monique Poulin and Rémy Pouliot for sharing their database of vegetation in natural Peatland of Eastern Canada and to Laurie Nadeau for field assistance.

References

- Amacher, M., Henderson, R., Breithaupt, M., Seale, C., LaBauve, J., 1990. Unbuffered and buffered salt methods for exchangeable cations and effective cation-exchange capacity. *Soil Sci. Soc. Am. J.* 54, 1036–1042.
- Andersen, R., Francez, A.-J., Rochefort, L., 2006. The physicochemical and microbial status of a restored bog in Québec: identification of relevant criteria to monitor success. *Soil Biol. Biochem.* 38, 1375–1387.
- Aronson, J., Clewell, A.F., Blignaut, J.N., Milton, S.J., 2006. Ecological restoration: a new frontier for nature conservation and economics. *J. Nat. Conserv.* 14, 135–139.
- Bellemare, M., Lapointe, L., Chiasson, G., Daigle, J.-Y., Rochefort, L., 2009. Conditions favouring survival of cloudberry (*Rubus chamaemorus*) rhizomes planted in cutover peatland. *Mires Peat* 5, 1–8, 7.
- Benayas, J.M.R., Bullock, J.M., Newton, A.C., 2008. Creating woodland islets to reconcile ecological restoration, conservation, and agricultural land use. *Front. Ecol. Environ.* 6, 329–336.
- Bennett, V.A., Doerr, V.A., Doerr, E.D., Manning, A.D., Lindenmayer, D.B., Yoon, H.J., 2013. Causes of reintroduction failure of the brown treecreeper: implications for ecosystem restoration. *Austral Ecol.* 38, 700–712.
- Blier-Langdeau, A., 2019. La réponse au feu des communautés végétales d'une tourbière ombrotrophe restaurée. MSc Thesis. Université Laval, Québec, Canada, p. 91p.
- Bray, R.H., Kurtz, L., 1945. Determination of total, organic, and available forms of phosphorus in soils. *Soil Sci.* 59, 39–46.
- Bunch, W.D., Cowden, C.C., Wurzbarger, N., Shefferson, R.P., 2013. Geography and soil chemistry drive the distribution of fungal associations in lady's slipper orchid, *Cypripedium acaule*. *Botany* 91, 850–856.
- Caporn, S.J.M., Rosenburgh, A.E., Keightley, A.T., Hinde, S.L., Riggs, J.L., Buckler, M., Wright, N.A., 2018. *Sphagnum* restoration on degraded blanket and raised bogs in the UK using micropropagated source material: a review of progress. *Mires Peat* 20, 1–17, 9.
- Coiffait-Gombault, C., Buisson, E., Dutoit, T., 2011. Hay transfer promotes establishment of Mediterranean steppe vegetation on soil disturbed by pipeline construction. *Restor. Ecol.* 19, 214–222.
- R Core Team, 2017. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria ([Online] R-project.org).
- Couwenberg, J., Thiele, A., Tanneberger, F., Augustin, J., Bährisch, S., Dubovik, D., et al., 2011. Assessing greenhouse gas emissions from peatlands using vegetation as a proxy. *Hydrobiologia* 674, 67–89.
- CSPMA, 2015. Statistics about Peatland Areas Managed for Horticultural Peat Harvesting in Canada. Canadian Sphagnum Peat Moss Association [On Line] peatmoss.com/. (Accessed 26 June 2018).
- Curtis, J.T., 1943. Germination and seedling development in five species of *Cypripedium* L. *Am. J. Bot.* 30, 199–206.
- Dalrymple, S.E., Banks, E., Stewart, G.B., Pullin, A.S., 2012. A meta-analysis of threatened plant reintroductions from across the globe. In: Maschinski, J., Haskins, K.E., Raven, P.H. (Eds.), *Plant Reintroduction in a Changing Climate. The Science and Practice of Ecological Restoration*. Island Press, Washington, USA, pp. 31–50.
- Deredec, A., Courchamp, F., 2007. Importance of the allee effect for reintroductions. *Ecoscience* 14, 440–451.
- Donath, T.W., Bissels, S., Hölzel, N., Otte, A., 2007. Large scale application of diaspore transfer with plant material in restoration practice—Impact of seed and microsite limitation. *Biol. Conserv.* 138, 224–234.
- D'Astous, A., 2012. Approches par communautés et par traits pour l'évaluation du succès de restauration d'une tourbière. Thesis. Université Laval, Québec, Canada, p. 88.
- D'Astous, A., Poulin, M., Aubin, I., Rochefort, L., 2013. Using functional diversity as an indicator of restoration success of a cut-over bog. *Ecol. Eng.* 61, 519–526.
- Fanelli, D., 2012. Negative results are disappearing from most disciplines and countries. *Scientometrics* 90, 891.
- Flora of North America (Ed.), 1993+. *Flora of North America*, 20+ (New York and Oxford).
- Fontaine, N., Poulin, M., Rochefort, L., 2007. Plant diversity associated with pools in natural and restored peatlands. *Mires Peat* 2, 1–17, 6.
- Gignac, D., Gauthier, R., Rochefort, L., Bubier, J., 2004. Distribution and habitat niches of 37 peatland *Cyperaceae* species across a broad geographic range in Canada. *Botany* 82, 1292–1313.
- Godefroid, S., Piazza, C., Rossi, G., Buord, S., Stevens, A.D., Aguraju, R., et al., 2011. How successful are plant species reintroductions? *Biol. Conserv.* 144, 672–682.
- González, E., Rochefort, L., 2014. Drivers of success in 53 cutover bogs restored by a moss layer transfer technique. *Ecol. Eng.* 68, 279–290.
- González, E., Rochefort, L., 2019. Declaring success in *Sphagnum* peatland restoration: identifying outcomes from readily measurable vegetation descriptors. *Mires Peat* 24, 1–16, 19.
- González, E., Rochefort, L., Boudreau, S., Hugron, S., Poulin, M., 2013. Can indicator species predict restoration outcomes early in the monitoring process? A case study with peatlands. *Ecol. Indic.* 32, 232–238.
- Graf, M.D., Rochefort, L., 2016. A conceptual framework for ecosystem restoration applied to industrial peatlands. In: Bonn, A., Allott, T., Evans, M., Joosten, H., Stoneman, R. (Eds.), *Peatland Restoration and Ecosystem Services: Science, Policy and Practice*. Ecological Reviews of Cambridge University Press, Cambridge, UK, pp. 192–212.
- Grey-Ross, R., Downs, C.T., Kirkman, K., 2009. Reintroduction failure of captive-bred oribi (*Ourebia ourebi*). *S. Afr. J. Wildl. Res.* 39, 34–38.
- Griffith, B., Scott, J.M., Carpenter, J.W., Reed, C., 1989. Translocation as a species conservation tool: status and strategy. *Science* 245, 477–480.
- Guéné-Nanchen, M., Hugron, S., Rochefort, L., 2018. Harvesting surface vegetation does not impede self-recovery of *Sphagnum* peatlands. *Restor. Ecol.* 27, 178–188.
- Hedberg, P., Kotowski, W., 2010. New nature by sowing? The current state of species introduction in grassland restoration, and the road ahead. *J. Nat. Conserv.* 18, 304–308.
- Hölzel, N., Otte, A., 2003. Restoration of a species-rich flood meadow by topsoil removal and diaspore transfer with plant material. *Appl. Veg. Sci.* 6, 131–140.
- Karofeld, E., Müür, M., Vellak, K., 2016. Factors affecting re-vegetation dynamics of experimentally restored extracted peatland in Estonia. *Environ. Sci. Pollut. Control Ser.* 23, 13706–13717.

- Keeney, D.R., Nelson, D.W., 1982. Nitrogen-Inorganic forms. In: Page, A.L., A.L., Miller, R.H., Keeney, D.R. (Eds.), *Methods of Soil Analysis—Part 2. Chemical and Microbiological Properties*, 2e ed., vol. 9. ASA-SSSA Madison, Wisconsin, pp. 643–698. Agron. No.
- Kiehl, K., Kirmer, A., Donath, T.W., Rasran, L., Hölzel, N., 2010. Species introduction in restoration projects—Evaluation of different techniques for the establishment of semi-natural grasslands in Central and Northwestern Europe. *Basic Appl. Ecol.* 11, 285–299.
- Koch, J.M., 2007. Restoring a jarrah forest understorey vegetation after bauxite mining in Western Australia. *Restor. Ecol.* 15, S26–S39.
- Laberge, V., Hugron, S., Poulin, M., Rochefort, L., 2015. Influence of different bryophyte carpets on vascular plant establishment around pools in restored peatlands. *Land Degrad. Dev.* 26, 813–818.
- Lady Bird Johnson Wildflower Center, 2014. Plant Database: *Maianthemum canadense*. The University of Texas at Austin [On Line] wildflower.org. (Accessed 14 March 2018).
- Landry, T., Rochefort, L., Poulin, M., 2012. Impact of seedbed and water level on the establishment of plant species associated with bog pools: implications for restoration. *Native Plants J.* 13, 205–215.
- Lefebvre-Ruel, S., Jutras, S., Campbell, D., Rochefort, L., 2019. Ecohydrological gradients and their restoration on the periphery of extracted peatlands. *Restor. Ecol.* 27, 782–792.
- Legendre, P., Legendre, L., 2012. *Numerical Ecology*, third ed., vol. 2. Elsevier, Oxford, UK, p. 1006.
- Lemay, M.A., De Vriendt, L., Pellerin, S., Poulin, M., 2015. *Ex situ* germination as a method for seed viability assessment in a peatland orchid, *Platanthera blephariglottis*. *Am. J. Bot.* 102, 390–395.
- Loreau, M., Naeem, S., Inchausti, P., Bengtsson, J., Grime, J.P., Hector, A., et al., 2001. Biodiversity and ecosystem functioning: current knowledge and future challenges. *Science* 294, 804–808.
- Luna, T., Wick, D., Hosokawa, J., 2008. Propagation protocol for production of container (plug) *Maianthemum stellatum* (L.) USDI NPS - glacier national park west glacier, Montana. In: Native Plant Network [Online] NativePlantNetwork.org. (Accessed 14 March 2018).
- McCarter, C.P.R., Price, J.S., 2015. The hydrology of the Bois-des-Bel peatland restoration: hydrophysical properties limiting connectivity between regenerated *Sphagnum* and remnant vacuum harvested peat deposit. *Ecohydrology* 8, 173–187.
- McDonald, T., Gann, G.D., Jonson, J., Dixon, K.W., 2016. *International Standards for the Practice of Ecological Restoration – Including Principles and Key Concepts*. Society for Ecological Restoration, Washington, USA, p. 47.
- Meli, P., Benayas, J.M.R., Balvanera, P., Ramos, M.M., 2014. Restoration enhances wetland biodiversity and ecosystem service supply, but results are context-dependent: a meta-analysis. *PloS One* 9, e93507.
- Nugent, K., Strachan, I.B., Strack, M., Roulet, N.T., Rochefort, L., 2018. Multi-year net ecosystem carbon balance of a restored peatland reveals a return to carbon sink. *Global Change Biol.* 24, 5751–5768.
- Oksanen, J., Blanchet, F.G., Kindt, R., Legendre, P., Minchin, P.R., O'hara, R.B., Simpson, G.L., Solymos, P., Stevens, M.H.H., Wagner, H., 2016. *Vegan: Community Ecology Package*. R package version 2.4-1. [On Line] cran.R-project.org/package=vegan.
- Paradis, É., Rochefort, L., 2012. Ecological restoration of lagg-swamp species on cutover peatlands. In: No 24. Proceedings of the 14th International Peat Congress: Peatland in Balance, Stockholm, Sweden, June 3–8, 2012. International Peat Society, p. 6.
- Paradis, É., Rochefort, L., Langlois, M., 2015. The lagg ecotone: an integrative part of bog ecosystems in North America. *Plant Ecol.* 2016, 999–1018.
- Perring, M.P., Standish, R.J., Price, J.N., Craig, M.D., Erickson, T.E., Ruthrof, K.X., et al., 2015. Advances in restoration ecology: rising to the challenges of the coming decades. *Ecosphere* 6, 1–25.
- Possingham, H.P., Bode, M., Klein, C.J., 2015. Optimal conservation outcomes require both restoration and protection. *PLoS Biol.* 13, e1002052.
- Poulin, M., Fontaine, N., Rochefort, L., 2011. Restoration of pool margin communities in cutover peatlands. *Aquat. Bot.* 94, 107–111.
- Poulin, M., Andersen, R., Rochefort, L., 2013. A new approach for tracking vegetation change after restoration: a case study with peatlands. *Restor. Ecol.* 21, 363–371.
- Quinty, F., 2005. Sainte-Marguerite-Marie peatland restoration project. Planirest Environnement. Saint-Charles-de-Bellechasse, Canada, p. 13.
- Quinty, F., Rochefort, L., 2003. *Peatland Restoration Guide*, second ed. Canadian Sphagnum Peat Moss Association and New Brunswick Department of Natural Resources and Energy, Québec, Canada, p. 106.
- Rochefort, L., Isselin-Nondedeu, F., Boudreau, S., Poulin, M., 2013. Comparing survey methods for monitoring vegetation change through time in a restored peatland. *Wetl. Ecol. Manag.* 21, 71–85.
- Society for Ecological Restoration International (SERI), 2004. *The SER International Primer on Ecological Restoration*. Society for Ecological Restoration International, Tucson [On Line] ser.org &.
- Shefferson, R.P., Weiss, M., Kull, T., Taylor, D.L., 2005. High specificity generally characterizes mycorrhizal association in rare lady's slipper orchids, genus *Cypripedium*. *Mol. Ecol.* 14, 613–626.
- Sjörs, H., 1950. On the relation between vegetation and electrolytes in north Swedish mire waters. *Oikos* 2, 241–257.
- Strack, M., Zuback, Y., 2013. Annual carbon balance of a peatland 10 yr following restoration. *Biogeosciences* 10, 2885–2896.
- Stravinsky, I., 2000. Writing the wrongs: developing a safe-fail culture in conservation. *Conserv. Biol.* 14, 1567–1568.
- Tilman, D., Isbell, F., Cowles, J.M., 2014. Biodiversity and ecosystem functioning. *Annu. Rev. Ecol. Syst.* 45, 471–493.
- Török, P., Vida, E., Deák, B., Lengyel, S., Tóthmérész, B., 2011. Grassland restoration on former croplands in Europe: an assessment of applicability of techniques and costs. *Biodivers. Conserv.* 20, 2311–2332.
- van Katwijk, M.M., Thorhaug, A., Marbà, N., Orth, R.J., Duarte, C.M., Kendrick, G.A., et al., 2016. Global analysis of seagrass restoration: the importance of large-scale planting. *J. Appl. Ecol.* 53, 567–578.
- Wheeler, G.A., Glaser, P.H., Gorham, E., Wetmore, C.M., Bowers, F.D., Janssens, J.A., 1983. Contributions to the flora of the Red Lake Peatland, northern Minnesota, with special attention to *Carex*. *Am. Midl. Nat.* 110, 62–96.
- Wiens, J.A., Hobbs, R.J., 2015. Integrating conservation and restoration in a changing world. *Bioscience* 65, 302–312.
- Wolters, M., Vries, S., Ozinga, W.A., Bakker, J.P., 2017. Restoration of inland brackish vegetation by large-scale transfer of coastal driftline material. *Appl. Veg. Sci.* 20, 641–650.